[illegible]

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9702/02

May/June 2008

1 hour

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen.

Do not use staples, paper clips, highlighters, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** questions.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
Total	

This document consists of **17** printed pages and **3** blank pages.

Data

speed of light in free space,

$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$

permeability of free space,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$$

permittivity of free space,

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$$

elementary charge,

$$e = 1.60 \times 10^{-19} \text{ C}$$

the Planck constant,

$$h = 6.63 \times 10^{-34} \text{ J s}$$

unified atomic mass constant,

$$u = 1.66 \times 10^{-27} \text{ kg}$$

rest mass of electron,

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

rest mass of proton,

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

molar gas constant,

$$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

the Avogadro constant,

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

the Boltzmann constant,

$$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

gravitational constant,

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

acceleration of free fall,

$$g = 9.81 \text{ m s}^{-2}$$

Formulae

uniformly accelerated motion,

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas,

$$W = p\Delta V$$

gravitational potential,

$$\phi = -\frac{Gm}{r}$$

hydrostatic pressure,

$$p = \rho gh$$

pressure of an ideal gas,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion,

$$a = -\omega^2 x$$

velocity of particle in s.h.m.,

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

electric potential,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series,

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel,

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor,

$$W = \frac{1}{2} QV$$

resistors in series,

$$R = R_1 + R_2 + \dots$$

resistors in parallel,

$$1/R = 1/R_1 + 1/R_2 + \dots$$

alternating current/voltage,

$$x = x_0 \sin \omega t$$

radioactive decay,

$$x = x_0 \exp(-\lambda t)$$

decay constant,

$$\lambda = \frac{0.693}{t_{1/2}}$$

Answer **all** the questions in the spaces provided.

For
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1 Make reasonable estimates of the following quantities.

(a) the frequency of an audible sound wave

frequency = Hz [1]

(b) the wavelength, in nm, of ultraviolet radiation

wavelength = nm [1]

(c) the mass of a plastic 30 cm ruler

mass = g [1]

(d) the density of air at atmospheric pressure

density = kg m^{-3} [1]

- 2 A spring is placed on a flat surface and different weights are placed on it, as shown in Fig. 2.1.

For
Examiner's
Use

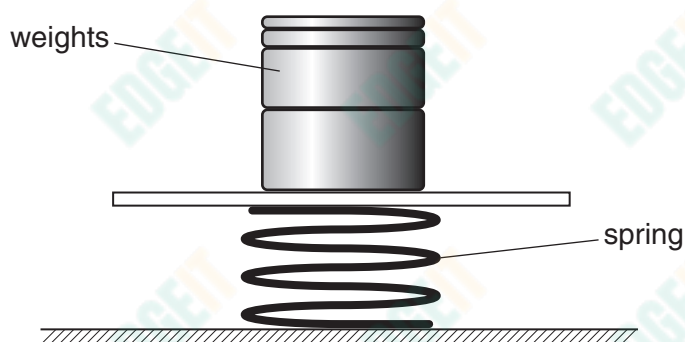


Fig. 2.1

The variation with weight of the compression of the spring is shown in Fig. 2.2.

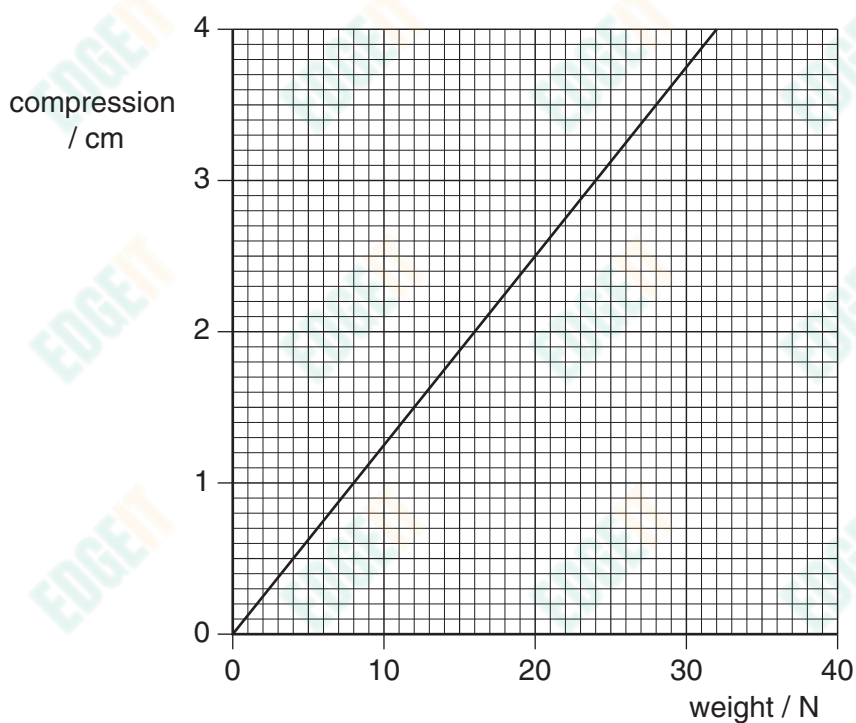


Fig. 2.2

The elastic limit of the spring has not been exceeded.

- (a) (i) Determine the spring constant k of the spring.

$k = \dots\dots\dots \text{Nm}^{-1}$ [2]

- (ii) Deduce that the strain energy stored in the spring is 0.49 J for a compression of 3.5 cm.

[2]

- (b) Two trolleys, of masses 800 g and 2400 g, are free to move on a horizontal table. The spring in (a) is placed between the trolleys and the trolleys are tied together using thread so that the compression of the spring is 3.5 cm, as shown in Fig. 2.3.

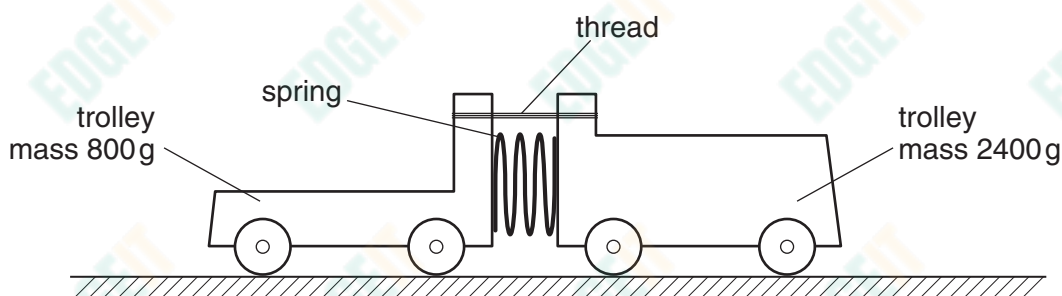


Fig. 2.3

Initially, the trolleys are not moving.
The thread is then cut and the trolleys move apart.

- (i) Deduce that the ratio

$$\frac{\text{speed of trolley of mass 800 g}}{\text{speed of trolley of mass 2400 g}}$$

is equal to 3.0.

[2]

- (ii) Use the answers in **(a)(ii)** and **(b)(i)** to calculate the speed of the trolley of mass 800 g.

For
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Use

speed = ms^{-1} [3]

- 3 A shopping trolley and its contents have a total mass of 42 kg. The trolley is being pushed along a horizontal surface at a speed of 1.2 m s^{-1} . When the trolley is released, it travels a distance of 1.9 m before coming to rest.

(a) Assuming that the total force opposing the motion of the trolley is constant,

(i) calculate the deceleration of the trolley,

deceleration = m s^{-2} [2]

(ii) show that the total force opposing the motion of the trolley is 16 N.

[1]

(b) Using the answer in (a)(ii), calculate the power required to overcome the total force opposing the motion of the trolley at a speed of 1.2 m s^{-1} .

power = W [2]

- (c) The trolley now moves down a straight slope that is inclined at an angle of 2.8° to the horizontal, as shown in Fig. 3.1.

For
Examiner's
Use

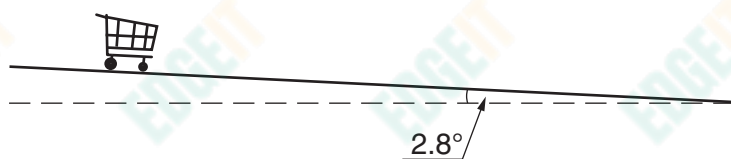


Fig. 3.1

The constant force that opposes the motion of the trolley is 16 N.

Calculate, for the trolley moving down the slope,

- (i) the component down the slope of the trolley's weight,

component of weight = N [2]

- (ii) the time for the trolley to travel from rest a distance of 3.5 m along the length of the slope.

time = s [4]

- (d) Use your answer to (c)(ii) to explain why, for safety reasons, the slope is not made any steeper.

.....
 [1]

4 (a) (i) Define the terms

1. tensile stress,

.....
[1]

2. tensile strain,

.....
[1]

3. the Young modulus.

.....
[1]

(ii) Suggest why the Young modulus is not used to describe the deformation of a liquid or a gas.

.....
[1]

(b) The change ΔV in the volume V of some water when the pressure on the water increases by Δp is given by the expression

$$\Delta p = 2.2 \times 10^9 \frac{\Delta V}{V},$$

where Δp is measured in pascal.

In many applications, water is assumed to be incompressible.
 By reference to the expression, justify this assumption.

.....

[2]

- (c) Normal atmospheric pressure is 1.01×10^5 Pa.

Divers in water of density $1.08 \times 10^3 \text{ kg m}^{-3}$ frequently use an approximation that every 10 m increase in depth of water is equivalent to one atmosphere increase in pressure. Determine the percentage error in this approximation.

For
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Use

error = % [3]

5 (a) State what is meant by

(i) the *frequency* of a progressive wave,

.....

 [2]

(ii) the *speed* of a progressive wave.

.....
 [1]

(b) One end of a long string is attached to an oscillator. The string passes over a frictionless pulley and is kept taut by means of a weight, as shown in Fig. 5.1.

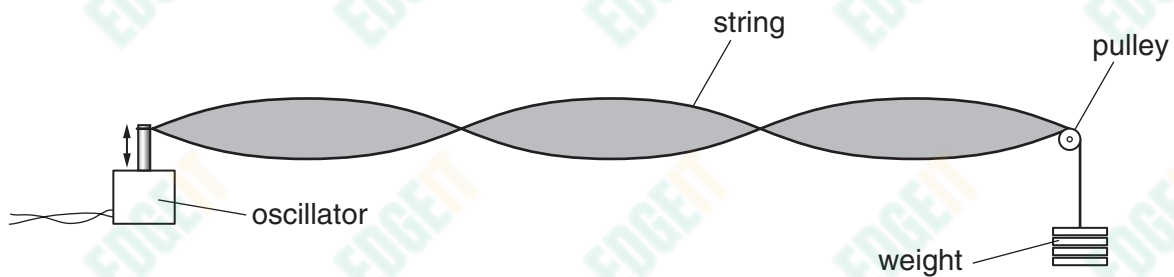


Fig. 5.1

The frequency of oscillation is varied and, at one value of frequency, the wave formed on the string is as shown in Fig. 5.1.

(i) Explain why the wave is said to be a *stationary wave*.

.....
 [1]

(ii) State what is meant by an *antinode*.

.....
 [1]

(iii) On Fig. 5.1, label the antinodes with the letter A.

[1]

- (c) A weight of 4.00 N is hung from the string in (b) and the frequency of oscillation is adjusted until a stationary wave is formed on the string. The separation of the antinodes on the string is 17.8 cm for a frequency of 125 Hz. The speed v of waves on a string is given by the expression

$$v = \sqrt{\frac{T}{m}},$$

where T is the tension in the string and m is its mass per unit length. Determine the mass per unit length of the string.

mass per unit length = kg m^{-1} [5]

- 6 An electric heater consists of three similar heating elements A, B and C, connected as shown in Fig. 6.1.

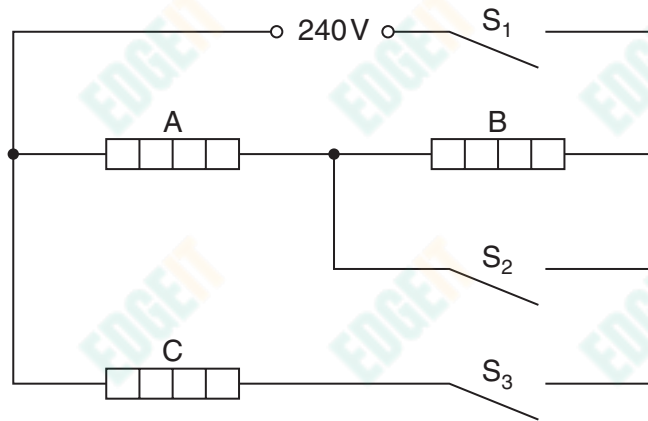


Fig. 6.1

Each heating element is rated as 1.5kW, 240V and may be assumed to have constant resistance.

The circuit is connected to a 240V supply.

- (a) Calculate the resistance of one heating element.

resistance = Ω [2]

- (b) The switches S_1 , S_2 and S_3 may be either open or closed.

Complete Fig. 6.2 to show the total power dissipation of the heater for the switches in the positions indicated.

For
Examiner's
Use

S_1	S_2	S_3	total power / kW
open	closed	closed	
closed	closed	open	
closed	closed	closed	
closed	open	open	
closed	open	closed	

[5]

Fig. 6.2

- (a)** Distinguish between an α -particle and a β -particle.

[4]

-
- A graph showing the number of neutrons N (y-axis) versus the proton number Z (x-axis) for Uranium isotopes. The y-axis ranges from 140 to 149, and the x-axis ranges from 88 to 97. Two points are plotted: $^{236}_{92}\text{U}$ at $(90, 142)$ and $^{238}_{92}\text{U}$ at $(92, 144)$. A line segment connects these two points, with an arrow pointing from $^{236}_{92}\text{U}$ to $^{238}_{92}\text{U}$.
- | Isotope | Proton Number (Z) | Neutron Number (N) |
|-----------------------|-----------------------|------------------------|
| $^{236}_{92}\text{U}$ | 90 | 142 |
| $^{238}_{92}\text{U}$ | 92 | 144 |

Fig. 7.1

The α -decay of Uranium-236 ($^{236}_{92}\text{U}$) is represented on the grid. This decay produces a nucleus of thorium (Th).

For
Examiner's
Use

- (i) Write down the nuclear equation for this α -decay.

.....[2]

- (ii) On Fig. 7.1, mark the position for a nucleus of

1. Uranium-237 (mark this position with the letter U),
2. Neptunium, the nucleus produced by the β -decay of Uranium-237 (mark this position with the letters Np). [2]

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